



Pacific Integrated Handling
10215 Portland Ave E
Tacoma, WA 98455

Presents to:

Thermo Fisher

SCIENTIFIC

Hillsboro, OR

Quote

Modula ML25 Vertical Lift Module



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Jonathon Johnson,

Pacific Integrated Handling thanks you for the opportunity to provide Thermo Fisher with the following automated storage solution for the cleanroom at the Hillsboro facility. The following proposal provides an in-depth description of the Modula Lift VLM we are proposing.

This Modula Vertical Lift Module (VLM) has been specified to meet Thermo Fisher's storage requirements and will aid in increasing storage density and capacity. Computerized management also enables all the logistics processes to be monitored and optimized. This vertical system will free up valuable floor space, minimize search time, and increase productivity utilizing the latest technology available.

The VLM will have a total storage capacity of 634.66 sq ft. and 363.6 cubic ft. of volume in a very small footprint of 124.28 sq. ft. While on-site, I completed a survey of the existing materials in the cleanroom and concluded that (with a 50% buffer) only 136.80 cubic ft is needed for storage. This will give Thermo Fisher plenty of room for growth or flexibility in what is kept within the VLM. A safety photo eye curtain system will protect operators during mechanical movements. All safety devices and machine designs meet CE standards. In addition, tool accessible doors on the side of the unit will allow service personnel access to the units for scheduled maintenance. The VLM will come with 2 years of parts and labor warranty as well as a *PIH exclusive 3 scheduled maintenance visits*.

Additionally, the VLM have a height of 145.67 inches (12' 2.64") and will contain 22 slotted trays - all of which are sized as 161.41" wide and 25.75" deep and will be capable of holding up to 551 lbs. Slotted trays will allow Thermo Fisher to use Modula dividers to create storage cells within the trays to organize smaller parts.

This proposal includes software installation of Modula WMS Premium, which is comprehensive software and will allow for the incorporation of bar code scanning, ERP integration, and future implementation of put-to-light systems, per the discussion on site.

The Modula VLM is considered ISO7-Ready and will be the ideal fit for the cleanroom environment at Thermo Fisher.

PIH has decades of experience with automated material handling systems. Our factory trained and certified technicians have a meticulous attention to detail and a reputation for the highest level of customer service and satisfaction, not only during the installation, but also on the following scheduled maintenance visits. Without hesitation I can say that PIH is the best partner for Thermo Fisher when it comes to your present and future automated high-density storage and retrieval system needs.

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Why VLMs?

- Significant Space Savings: VLMs free up more space for further storage or revenue-generating equipment.
- Height Ergonomics: VLM delivery heights increase safety and reduce injuries, therefore reducing workers' comp claims. VLMs eliminate the need for a human to step up on a stool or stoop over to pick parts from lower shelves.
- VLMs reduce search times: Heightened organization within the VLM will allow pickers to find their needed tools and materials quickly and efficiently – allowing for increased time spent working on production-related activities vs time searching.
- VLMs will decrease the chance of damaged or stolen parts.
- VLMs make maximum usage of facility height: Existing shelving doesn't make use of vertical space as efficiently and takes up a much larger footprint than a VLM.
- VLMs require significantly less training for picking: Using a VLM sheds the need for tribal knowledge and allows for quick and seamless obtaining of materials.
- VLMs have 24/7 traceability of inventory management (who is picking items and when?).
- VLMs are far denser and more configurable than drawers: drawers are never 100% optimized and typically have a 20-30% usage rate.
- VLMs provide secure storage of all materials.
- The aesthetic of the VLMs is very modern compared to shelving.
- Stock Density Optimization: A height-detection system determines vertical dimensions of items as the tray is put away, calculating adjustments in real time to maximize storage density within the unit.

Why Modula?

- Components of the Modula VLM are off the shelf and not proprietary (greater availability at lower cost).
- One-time cost for software. Includes software version updates at no additional cost.
- VLM control panel (co-pilot) is at an ergonomic height and in a location that minimizes risk of safety barrier tripping.
- Modula has the most advanced anti-seismic system on the market today.
- Modula WMS has a highly intuitive, modern user interface.
- Modula VLMs are the only VLMs which can meet "Made in USA" and "Buy America" requirements.

- VLM software can interface with Corridor or any other ERP/tool management system to ensure accountability of items both inside and outside the machine. It also can operate as a robust, standalone system.

Why Pacific Integrated Handling (PIH)?

- PIH is the Tier 1 dealer for Modula in the Pacific Northwest, which comes with the following exclusive benefits:
 - o PIH boasts our own team of factory-trained service and support technicians.
 - o Certified Technicians are all outfitted with fully stocked service vans.
 - o Direct partnership and engagement with the Modula support team.
- **PIH offers 3 scheduled maintenance visits throughout the 2-year warranty period.**
 - o **Modula and their partners only require and perform 1 visit.**
- PIH is the largest Modula VLM dealer in the PNW, actively servicing dozens of clients across Oregon and the rest of the Western seaboard.
 - o This high volume of VLMs under PIH service agreements directly results in highly trained and experienced technicians.
- PIH offers a full turnkey solution without the need for outsourcing labor.
- PIH stocks over \$250,000 worth of spare parts in case of mechanical emergencies with our valued customers.
- PIH's Systems Division is singularly focused on high density automated storage and retrieval systems and has been since 1981.
- PIH has a fully staffed CAD design team dedicated to supporting the conceptualization and installation of VLMs in our customers' facilities.

MODULA®LIFT VLM specifications

Unit information:	MODULA®LIFT VLM Model:		ML25 – 3700			
Unit dimensions:	Unit height:		145.67	in.	3700	mm
	Unit width:		177.83	in.	4517	mm
	Unit depth:		100.63	in.	2556	mm
	Required clear ceiling height:		Unit Height, Plus 3.94 in. (100 mm)			
Trays:	Trays included per unit:		22			
	Tray sidewall type:		Slotted		Tray side walls	
	Usable tray width:		161.41	in.	4,100	mm
	Usable tray depth:		25.75	in.	654	mm
	Tray side wall height:		4.72	in.	120	mm
	Tray storage adjustability:		0.98	in.	25	mm
	Loads / Unit capacities:	Max. tray load capacity:		551	lbs.	250
Total max. machine weight as configured:		20557	lbs.	9323	kg.	
Net unit load capacity (as configured): (No. trays included x max. tray load capacity)		12,125	lbs.	5,500	kg.	
Empty unit weight w/ 22 Empty trays:		8,428	lbs.	3823	kg.	
Empty tray weight (per tray):		186.29	lbs.	84.50	kg.	
Power requirements:	Total installed power consumption:		7.6	kW		
	Power supply required:		400-480	V AC / 3~ / 60 Hz / w/o Neutral		
	Amperage draw / breaker size required:		12	Amp		
Customizations:	RAL grey:		☐			
	Flat panels:		☐			

PICKING BAY CONFIGURATION (PER UNIT):

Delivery locations:	Bay location:	Picking bay type:		Tray delivery heights:			
First:	Front	Internal		27.56	in.	700.00	mm
		Single	Delivery	0.00		0.00	
Second:				0.00	in.	0.00	mm
			Delivery	0.00		0.00	
Third:				0.00	in.	0.00	mm
			Delivery	0.00		0.00	

TRAY CONFIGURATIONS (PER UNIT):

Trays:			Usable height per tray:		Tray quantities:			Square area stored (ft ²)		Cubic Vol. stored (ft ³)	
					Max. qty.	Actual qty.	Qty. slotted	Max.	Offered	Max.	Offered
4.72	in.	Sidewalls	4.72	in.	7	7	7	201.94	201.94	79.51	79.51
4.72	in.	Sidewalls	6.69	in.	6	6	6	173.09	173.09	96.47	96.47
4.72	in.	Sidewalls	8.66	in.	9	9	9	259.63	259.63	187.63	187.63
	in.	Sidewalls		in.	0	0	0	0.00	0.00		0.00
	in.	Sidewalls		in.	0	0	0		0		0.00
	in.	Sidewalls		in.	0	0	0	0.00	0.00		0.00
TOTAL per unit:					22	22	22	634.66	634.66	363.6	363.65

UNIT PERFORMANCE AS CONFIGURED (PER UNIT):

		No Load:	Full Load:
Average tray delivery time:		18 sec.	21 sec.
Maximum tray delivery time:		22 sec.	25 sec.
Number of trays delivered per hour:	Picking time = 0 sec.	98	87
	Picking time = 15 sec.	70	64
	Picking time = 30 sec.	54	51

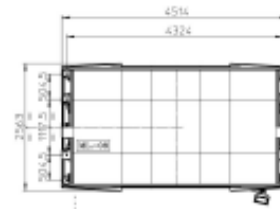
Modula Lift ML25 Specifications

Unit Dimensions

Modula Lift Model ML25 with Internal Bay Work Station



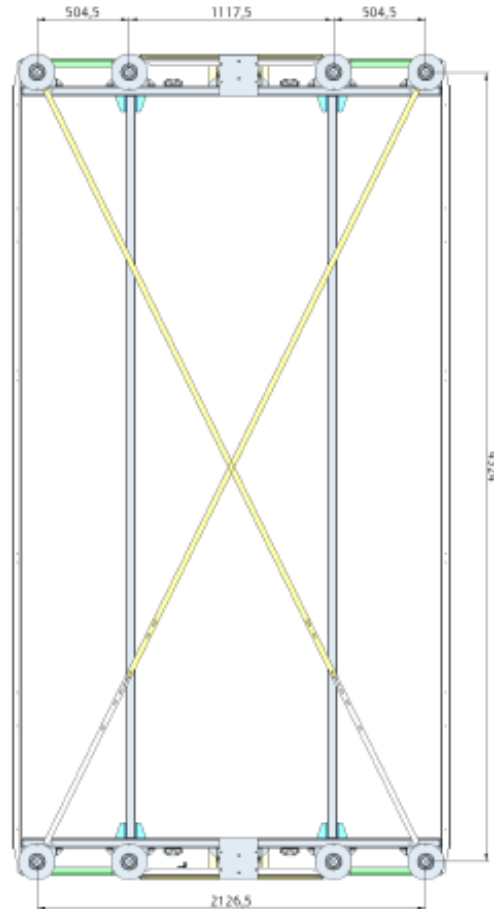
	Inches	mm
Unit Height:	145.67	3,700
Unit Width:	177.83	4,517
Unit Depth:	100.63	2,556
Floor Level Bay 1:	0.00	0
Floor Level Bay 2:	N/A	N/A
Floor Level Bay 3:	N/A	N/A



Power Requirements:

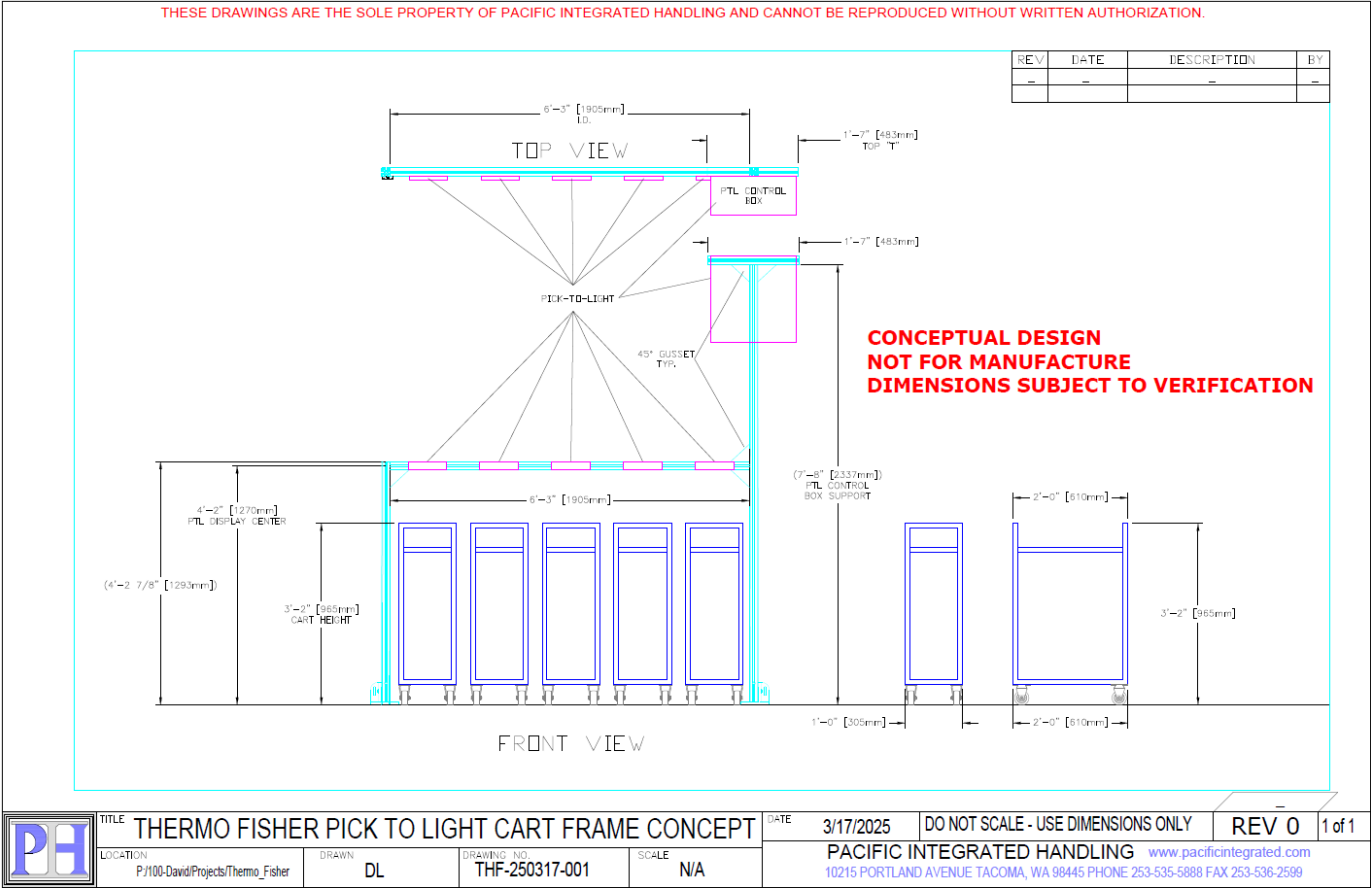
Total Installed Power Consumption:	6,5	kW
Power Supply Required:	208	V AC / 3~
Frequency:	50/60	Hz (w/o) Neutral
Main Power Disconnect Switch Type / Sizing:	300	mA - Class B Differential Switch

Modula Lift Weights and Loads



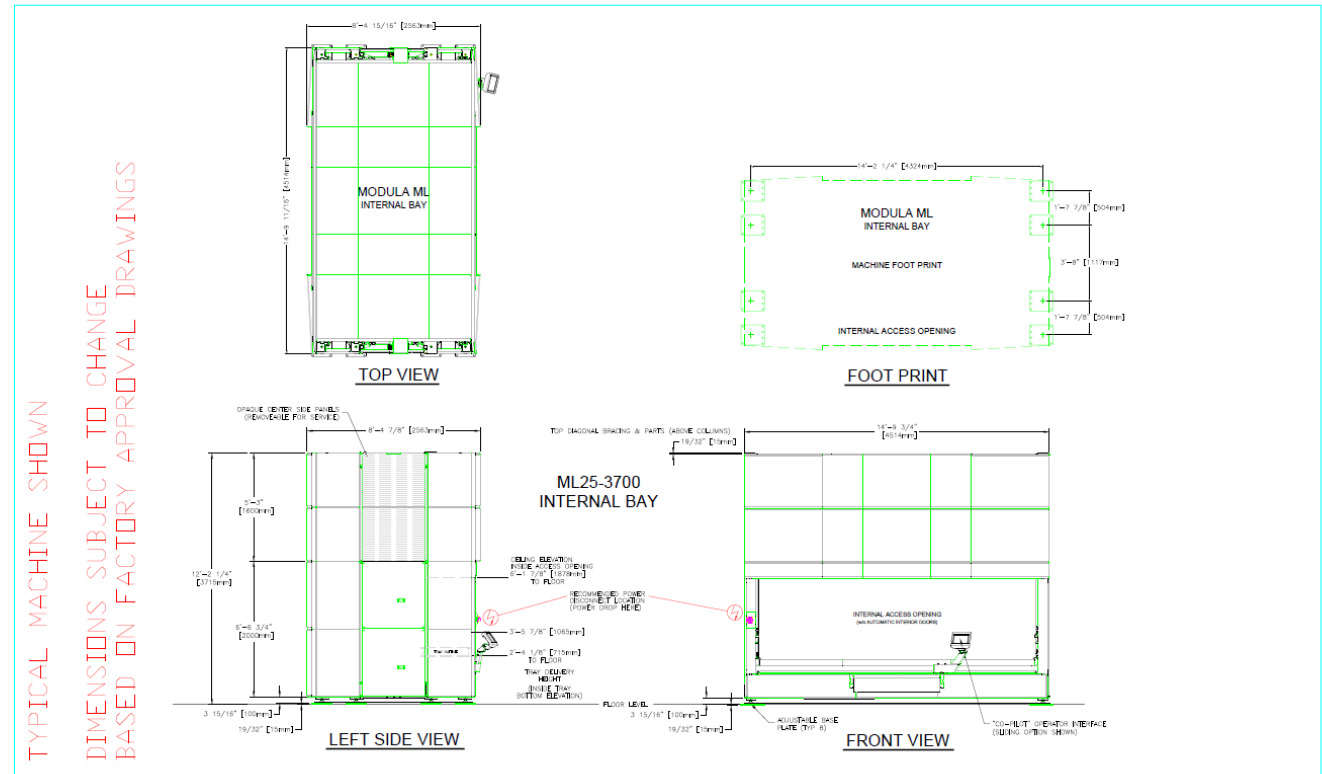
Machine structure weight, empty	1964	kg	4331	lbs
Weight of 22 trays, empty	1859	kg	4099	lbs
Weight of machine structure + 22 empty tray	3823	kg	8430	lbs
Maximum net payload capacity per tray	250	kg	551	lbs
Total net payload capacity with 22 trays	5500	kg	12128	lbs
Total maximum weight of machine as configured	9323	kg	20557	lbs
Floor area of machine	11,55	m ²	124.32	ft ²
Machine area average floor loading as configured	807	kg/m ²	165	lbs/ft ²
Surface area of each base plate	969,98	cm ²	150.35	in ²
Number of machine base plates	8	nr	8	ea
Base plate surface pressure as configured	1,20	kg/cm ²	17.09	lbs/in

Put to Light Concept (Not Curently Included in Price, also sent as separate attachment)



Modula ML25-3700 Drawings (also sent as separate attachment)

THESE DRAWINGS ARE THE SOLE PROPERTY OF PACIFIC INTEGRATED HANDLING AND CANNOT BE REPRODUCED WITHOUT WRITTEN AUTHORIZATION.



	TITLE MODULA ML25-3700 VLM WITH INTERNAL TRAY DELIVERY	DATE 3/17/2025	DO NOT SCALE - USE DIMENSIONS ONLY	1 of 1
	LOCATION P/100-David/Machine Drawings	DRAWN DL	DRAWING NO. ML25-3700_INT	SCALE N/A
	PACIFIC INTEGRATED HANDLING www.pacificintegrated.com			
	10215 PORTLAND AVENUE TACOMA, WA 98445 PHONE 253-535-5888 FAX 253-536-2599			

System Investment: 1 Modula ML25 VLM

Unit Pricing			
Items	Price Ea.	Qty.	Prices
MODULA® ML50-3700 VLM With Single-Tray Presentation	\$128,229.00	1	\$128,229.00
ISO7 Cleanroom Ready	Included	1	Included
Single Bay & Internal WorkStation	Included	1	Included
Tray size: 161.41" W x 25.75" D Tray Height 4.72"	Included	22	Included
Dynamic Height Management System	Included	1	Included
Co-Pilot Touchscreen Operator Console	Included	1	Included
Automatic Closing Door	Included	1	Included
Laser Pointer (Picking Aid)	Included	1	Included
Alphanumeric LED Display Bar (Picking Aid)	Included	1	Included
Telephone Support – 8 Hours/Day, 5 Days/Week – 2 Years	Included	1 lot	Included
Freight to Hillsboro, OR (1 Truck, Rear Load)	Included	1 lot	Included
Software			
Modula WMS Premium Software (one-time fee, includes updates)	\$18,813.00	1	\$18,813.00
Modula Advanced Picking Module	Included	1	Included
Modula On-Site Installation and Training	Included	1	Included
PIH On-Site Software Support	Included	1	Included
Installation and Warranty			
Mechanical Installation (includes rentals)	\$24,400.00	1	\$24,400.00
PIH Project Management	Included	1 lot	Included
Parts and Labor Warranty (2 Years)	Included	1 lot	Included
Seismic and Anchoring Calculations and Certification	\$5,347.00	1 lot	\$5,347.00
<i>PIH Exclusive: 3 Scheduled Maintenance Visits</i>	Included	1 lot	Included
Total:			\$176,789.00



Alphanumeric LED Bar



Sliding Console



Laser Pointer

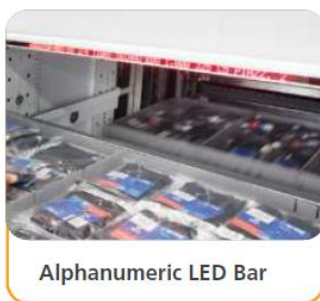
Options Not Included	Price Ea.	Qty.	Prices
ESD Protection System	\$1,200.00	1	Option
Sliding Operator Condole	\$1,535.00	1	Option
Wireless Bar Code Reader (1D)	\$1,793.00	1	Option
Wireless Bar Code Reader (2D)	\$3,000.00	1	Option
RFID Badge Reader	\$1,013.00	1	Option
Magnetic Badge Reader	\$607.00	1	Option
Spare Parts Kit – Level A	\$1,818.00	1	Option
Spare Parts Kit – Level B	\$9,142.00	1	Option
Spare Parts Kit – Level C	\$15,848.00	1	Option
Premium 24/5 Telephone Support Agreement (Annual Fee)	\$4,515.00	1	Option
Premium 24/7 Telephone Support Agreement (Annual Fee)	\$7,449.00	1	Option
Tray Partitions and Dividers	TBD	1	Option
Put-to-light System	TBD	1	Option

Note: Items marked in bold on “Options Not Included” are highly recommended by PIH for this project.

Lead Time Currently 10-12 weeks

PRICING IS SUBJECT TO FINAL CONFIGURATION APPROVAL

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Proposal includes:

- All mechanical hardware as outlined in this proposal
- Freight to Hillsboro, OR
- Unloading the equipment upon arrival
- Moving equipment from dock area to installation area
- Mechanical installation using our own forklift and scissor lifts
- Electrical installation and controls within the equipment
- Configuring controls and checking the equipment for proper operation
- Operator and maintenance training
- Manuals and Documentation
- PIH Exclusive: 3 Scheduled Maintenance Visits over the first 2 years (Modula requires 1)
- 2 Year Warranty Parts and Labor

Proposal does not include:

- Electrical Hookup to VLM provided by Thermo Fisher
- Fire Suppression
- Installation Permits
- Providing a clear path for moving equipment to the installation area
- Providing a clear area for installation and the erected equipment
- Computer Hardware and data drops
- Local and State Taxes on applicable items
- Seismic reports may require modifications to the facility which will be the customer's responsibility
- Dumpster Rental
- Partitions and Dividers to be discussed

Payment Schedule:

35% due upon purchase order, Net 10 days
55% due upon shipping, Net 30 days
10% due upon acceptance, Net 30 days

- Note 1. All Permits are "By Customer". Please note that processing time can take 3 to 5 weeks or more. This processing does NOT include the actual Permit cost which will be billed at cost.
- Note 2. Seismic calculations for the equipment are required to obtain installation permits. Seismic calculations will be prepared by an Architect selected by Pacific Integrated Handling. The purpose of the calculations is to show that the floor can support the installed storage equipment. The Architect will prepare final project calculations and drawings based on customer supplied floor data. The data needed includes floor hardness and thickness, rebar size and spacing, and the soil compaction or floor load rating and construction if not on ground floor. If the floor proves to be inadequate to support the equipment, the cost of changes to the floor and/or equipment is the responsibility of the customer.

Best regards,

Josh Jancola, Sales Representative

Accepted by: _____
Signature: _____
Date: _____

MODULA® LIFT VLM Features



Modular Frame Construction

Units feature a modular frame construction, being fully structural and not relying on exterior cladding to add to the unit's strength. Unit heights can easily be increased or decreased as needed in the future.

Steel Reinforced Toothed Belt Lifting Design

Heavy-duty, steel reinforced toothed lifting belts provide unsurpassed safety with secure lifting on both sides of the tray.

Tray Weight Monitoring

Units measure and log the tray's product load each time a tray is stored and automatically protects from the possibility of overloading.

Single-Level or Dual-Level Tray Delivery Types

Trays may be delivered to the user at either a single lower level, or when required, the second tray in queue is ready and delivered above the lower after picking is complete.

WMS Software

WMS Software with a wide range of functionalities available allows for simple integration to your HOST system and provides rich, simple information to users.



Internal or External Picking Bay Types

Internal-bay configurations available for smaller parts storage and reduced unit footprints. An industry exclusive external-bay configuration is also available for the fastest throughput rates and true top picking application.

Automatic Closing Door

An automatic shutter type door for the access opening area is available. The door provides increased security, aesthetics, and sound reduction.

User Safety

Units comply to strict operator safety standards. Controls and electronics comply with UL standards.

Dynamic Product Height Measurement & Storage

Products located in trays are measured precisely each time stored, ensuring the highest storage density possible with 1.0" adjustability.

CoPilot Operator Interface

The 10.5" color touch screen interface provides full access & information to the user from any task required, including graphical tray layouts.

CoPilot® Operator Interface Features

Integrated Emergency Stop (E-Stop) Button

An emergency stop (E-Stop) button is located on the top of the CoPilot® controller and is highly visible. Due to the E-Stop button's placement it is easy to reach and always near the operator.

Operator Login With Password

Allows users restricted access to different levels depending on their role(s):

- **Operator:** Basic usage of the machine, call by tray, part, etc.
- **Advanced:** Able to perform all the above, plus access log file information, run advanced reports, run compaction, etc.
- **Maintenance:** Able to use manual machine controls and make changes to machine parameters, etc.

Graphical Tray Layout

A graphical representation of the tray shows the operator precisely where to pick. Each cell is color coded to indicate the location is by available current quantity.

Easy To Use LIFT OS Operator Interface

Icon based environment; easiest most intuitive HMI in the market. Items can be picked by the tray number or by each unique item number. Photos of the individual item can be programmed to be displayed.

Displays Useful Tray Information

Useful machine information is clearly displayed such as, tray number at the access opening, the trays current loaded weight, and the trays current height occupied by product.

Industrial 10.5" Color Touch Screen Display

The 10.5" color touch screen is 50% larger when compared to the competition. The IP-67 rated industrial controls enclosure protects against particulates, water, and is suited for industrial environments.

Operate In Multiple Modes

- **Manual Mode:** Allows technicians to manually operation the unit for service work and adjustment.
- **Automatic Mode:** Allows for standard machine operations, picking by tray, item number, etc.
- **Remote / Host Mode:** Integration & communication to ERP systems via standard Ethernet connection

Displays Clear, Detailed Information

Information is clearly presented to the operator displaying all required information such as: item number, qty. to pick or put, order number, and item notes. Operators are also able to modify orders directly from the controller as well as print labels for orders.

Complete Machine Diagnostics Onboard

An integrated unit run-time program logs machine usage and alerts users when preventative maintenance is required. Simple icon based text and visuals guide users to help correct faults or errors. Real-time applicable machine information is displayed quickly and clearly.

USB Port Integrated Into The Controller

The integrated USB Port allows the user to connect optional peripheral devices such as barcode readers, badge readers, label printers, and EKS readers. It also allows easy diagnostics and updates.



Features That Make The MODULA® LIFT VLM Unique

What Makes the MODULA® VLM Unique?

CoPilot® Controller:

- Large, 10.5" color touch screen.
- Industrial Windows® based industrial PC, IP65 rated enclosure for rugged, industrial environments.
- Graphically indicated item picking and replenishment, for operator ease of use.
- Icon based OS environment, highly intuitive and easy to use for operators.
- 100% control of software at the CoPilot® unit controller.



Operator Interface Location:

- The CoPilot® controller is located at a convenient, ergonomic height and location at the picking bay area.
- Eliminates any light curtain nuisance faults caused by operators reaching for the controller on other units where located overhead inside the picking bay.
- Standard sliding option keeps control near the operator at all times.
- Integrated E-Stop, located conveniently on the controller has it nearby at all times.



Access Opening Lighting on Internal Bay Units:

- Provides for a well-lit picking bay area, aiding operators in clearly seeing what they are picking from the tray.
- The access opening lighting turns off / on automatically to clearly indicate to the operator when it is safe to pick, so that nuisance light curtain faults are greatly minimized.
- When it is safe to pick from the tray, the lights turn and stay on. When the pick is complete and the next tray is being positioned, the lights turn off, showing the operator that they should not be picking at that time.



Dual Tray Delivery (Continuous Pick Feature):

- True dual-delivery provides greatly increased operator productivity.
- The next tray in the picking queue is presented to the operator in seconds after the previous tray picks are complete.
- This nearly simultaneous tray exchange provides for up to a 33% increase in picking productivity.
- Greatly reduced idle time for operators.



Internal & External Bay Type Configurations:

- The available Internal Picking Bay provides operators with the most head room clear space within the picking bay area vs. competition for improved ergonomics.
- All units are available with an automatic opening picking bay door. The door features superior single piece construction for enhanced durability and security.
- The available External Picking Bay provides true overhead picking, increased unit storage capacity and increased throughput rate when compared to traditional, competitive internal bay systems.



External Picking Bay:

- The External Picking Bay provides the highest available throughput rate when compared to traditional, competitive internal bay systems.
- Trays and product are automatically delivered out of the unit to the external bay area quickly seamlessly with no additional steps required, providing true top picking to goods stored within.
- Ideal for applications requiring the use of an overhead lifting device for loading or picking from trays.
- The picking bay area ensures a safe and ergonomic working area for operators.



Telescopic Internal Picking Bay:

- Allowing an operator to temporarily extend the surface of the internal bay area, out of the unit for top picking where a lifting aide device is required or if product is particularly heavy or large.
- The Telescopic Internal Picking Bay is the ideal solution for applications requiring occasional overhead picking or loading needs.
- As the tray is able to easily extend from within the picking bay when needed, the required unit footprint is reduced.



- [illegible]

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- A computer monitor is shown in a laboratory setting, displaying a Gantt chart software interface. The screen shows a project schedule with various colored bars (yellow, pink, blue) representing different tasks or activities over time. The interface includes a header with project details and a main area with a timeline and task list. The monitor is on a desk, and there are some papers and a small green object (possibly a USB drive) in front of it. In the background, there are laboratory shelves and equipment.

Standard Features of the MODULA® LIFT VLM

Safety Features :

All MODULA® LIFT VLM units come standard with the following safety features :

- Personnel and product protection photo eyes, located in the workstation.
- Emergency stop button, located on the CoPilot® operator console / HMI.
- Automatic machine diagnostics on machine start-up.
- Tray weight monitoring will reject movement of the tray if it is overloaded.
- Front and rear tray stops hold trays securely in place at all times.

Dynamic Tray Height Monitoring System :

Product heights are automatically measured by the machine's on-board sensors and controller during tray insertion and the tray is then assigned to the optimal location within the machine.

Trays storage adjustability is provided on 1.0" increments of vertical spacing, allowing for the most efficient use of space possible. Also the unit's controller allows for the ability of the trays to be automatically compacted, optimizing storage capacity within the unit.

Lifting & Tray Stability Mechanisms :

Vertical tray movement is accomplished by a toothed belt drive and a self-braking gear motor controlled by an inverter. Vertical positioning is controlled by a count encoder.

Tray stability is provided by sixteen (16) rollers on the elevator, which guide along the unit's structure.

This lifting system removes the need for troublesome chain drives or rack and pinion designs, both of which require more maintenance, are louder while operating, and are more costly to repair.



Electrical Cabinet :

- Conveniently located under the picking bay area.
- The cabinet is mounted on rails for easy roll-out to access internal electronic components.
- The main electrical cabinet of the MODULA® LIFT VLM has been certified and features CE/UL listed components.

MODULA® LIFT VLM Optional Configurations & Accessories

Foot bar task complete



The device is installed on the ground in front of the bay and is activated by pressing an earth bar. Thus, the operator confirms “end picking” even when his hands are full, restricting the use of the Copilot to the minimum. Very comfortable and practical when the operator is holding goods or boxes and makes his job smoother and quicker.

Tray Transportation Cart



The optional Tray Removal and Transport Cart allows users to take fully loaded trays from within the VLM unit and bring them to another area for the applications requirements.

RFID Reader



The RFID reader acts as a swipe Badge reader, but also has another type of reader. There is no need to swipe with the RFID reader. Just place the badge close to the reader and authentication is wireless and contactless.

Robot Interface



The Modula storage system can be interfaced with anthropomorphic robots to pick up and drop-off. Two preparations are required:

- For the hardware: 5 signals on the electric control panel, 3 of which from Modula to the robots and 2 from the robot to Modula to indicate the operational levels and the robot's position compared to Modula.
- For the software: a specific protocol in the WMS Premium to communicate with the robots. This is a communication via TCP/IP sockets with data exchange on PLC memory areas controlled by the robot.

Telescopic Internal Bay



The available Telescopic Internal Bay configuration is available on all Internal Bay units. The Telescopic Internal Bay allows users to quickly and easily pull a tray out, for easy loading or picking of items using an overhead lifting device.

Laser Pointer



This is a windage pointer on 2 axes. The picking point is viewed through the intersection of a “red” holographic laser line, that identifies the item position on the “X” axis (width) and a “green” laser spot light which identifies the position on axis “Y” (depth). The pointer connection to the Modula system is ensured by an Ethernet type data connection.

Alphanumeric LED Bar



The optional Alphanumeric LED Bar clearly displays information to operators, indicating item position, description, quantity to pick, item number or any additional notes desired to be displayed. The result is increased picking accuracy and speed as operators are quickly and easily able to locate the correct items.

X-Axis LED Bar



The optional X-Axis LED Bar clearly displays the item's position information to operators. The result is increased picking accuracy and speed as operators are quickly and easily able to locate the correct items.

Sliding CoPilot® Console



The optional Sliding CoPilot Console allows operators to have the CoPilot with them at their picking location at all times. This eliminates the need to walk back to the controller to confirm operations or to select the next tray or item to request.

Tray Accessories



For a better and clearer arrangement of materials inside the trays, you may use partitions and dividers: the trays may be subdivided into compartments of various sizes. The tray layout can be changed when required, based on customer's needs, simply by moving the above mentioned accessories. Furthermore, the height of the tray side panels can be increased by means of the wall extensions.

ESD Protection System



This protection system has been designed to safeguard the components subject to electrostatic discharges. The ESD kit includes one antistatic wrist strap featuring an intrinsic resistance of 1MΩ. Additional wrist straps and the antistatic mat may be supplied on request as optional items.

Put-To-Light



The optional Put-To-Light System allows the operator to increase productivity, by being able to pick a higher number of orders at the same time, commonly referred to as batch picking. The result is greatly increased order throughput for the operator.

 Badge Reader



The USB badge reader, which can be connected directly to the operator's control console via the USB port, is the ideal tool when further control levels over user logins are desired. The operators use their own personal badges to access the machine, thus preventing unauthorized accesses in the event that the standard login procedure with password entry is not sufficient.

 EKS Reader



EKS is an electronic key reader which may be programmed with a dedicated SW. The electronic keys are equipped with an internal tag consisting of a memory chip and a transponder antenna, and are therefore battery-free. This device control also log-outs automatically, when the electronic key is removed from the reader.

 Bar Code Reader



The optional Bar Code Reader allows operators to scan items as they are being picked or replenished for item verification. Operators may also log into the machine by scanning their unique badge bar code. The Bar Code Reader is available in 1D and 2D versions to meet application requirements.

 Label Printer



The optional Label Printer quickly and easily prints labels for items and/or their shipping containers at the VLM workstation area. Available in multiple models depending on label type requirements to meet an applications needs.

 Piece Counting Scale



This device can count the number of items placed on the scale platform based on the unit weight of each individual piece; it is the ideal tool when large quantities of small size components are stored. The scales may come with a single platform with a load capacity of 3/6/15/30 kgs or with double platform, used to weigh items with very different weights.

 MODULA® Green



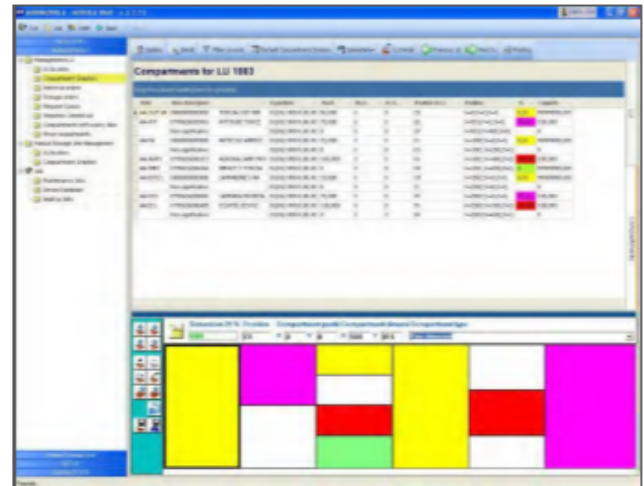
This device allows the user to recover the energy generated by the elevators downward movement, which would be otherwise wasted through heat. A special inverter fitted between the machine switch and the electric panel acts as both a voltage stabilizer and generator. This system leads to increased respect for the environment as it provides up to 40% energy saving.

MODULA® WMS Software

MODULA® WMS Software Features

MODULA® WMS is a complete inventory management software for all MODULA® automated storage equipment. The software is built on a relational database using Microsoft® SQL Server Express 2012. The interface is designed in the Microsoft® .Net environment.

- The standard use of a modern relational database on all software modules.
- Easy and fast data backup as well as data restoration in the event of a software malfunction.
- A customizable grid / screen layout to meet the needs of individual users.
- Advanced filtering options allowing for customizable searches and orders.
- Advanced user / operator rights management.
- Allows for the use of RF terminals for manual storage area management.
- Hierarchical item list management (Kitting).
- Graphical location management for the VLM unit(s) as well as manual areas (*shelving, rack, etc.*).
- Configurable integration with enterprise management systems via an ASCII file exchange. Data exchange on shared tables is also accessible via ODBC driver, XML files, and Excel spreadsheets.
- Packing list management.
- Material transit management.
- The ability to batch pick items picked from different locations, as well as the ability to utilize pick-to-light / put-to-light displays.
- Internal material movement management between different areas based on minimum stock values defined for each area.
- The ability to create custom reports with an easy to use, intuitive report tool.
- Automatic database maintenance to optimize the software's performance.



Software Architecture

MODULA® WMS's architecture is based on a PC (*or server PC*) containing the SQL database installed on the MODULA® master PC, e.g. the main program that allows the management of the VLM(s) and all the communication interfaces between the database and the host system and the machines and that also provides the configurable automatic backup. The PC must be provided with two network cards, one dedicated to the MODULA® sub network and one connected with the enterprise network. It is possible to install the software on up to two (2) additional client PC's also residing within the local network. Additional licenses are available in groups of three (3) as an option.

MODULA[®] WMS Software Modules Available

MODULA[®] WMS - BASE



The MODULA[®] WMS – BASE software module, is the **Included** package provided with MODULA[®] LIFT VLM units.

It provides some basic tools for vertical warehouse management, such as order register management, graphical tray location management, requesting items by means of orders or direct item requests, execution of physical inventories, and integration with the client's management system (host) by means of manual text file exchange with a predefined path.

The available features of the MODULA[®] WMS – BASE software module are:

- Item list management allowing operators to pick by part number.
- The ability to manage one or more MODULA[®] LIFT VLM units.
- VLM location management with the use of an easy to use graphical tool.
- Item movement management for the picking and replenishment of orders.
- Item movement management with the ability to perform immediate requests (*hot-picks*).
- User / operator management based on three distinct hierarchical access levels, depending on their role.
- The ability to manually import / export data to and from with the enterprise host system via a fixed – layout ASCII file exchange.

MODULA[®] WMS – AUTOMOTIVE



The MODULA[®] WMS – AUTOMOTIVE software module adds two important software functionalities in addition to the MODULA[®] WMS – BASE software module.

- Advanced security management, providing for a user's database with personalized operator's passwords, beneficial to users that desire to lock out specific trays for certain operators / users, for example.
- An automatic and configurable import / export data exchange with a client's management system (host), using text files or shared tables accessible via ODBC, .XML files, ASCII files, or using Excel files.

MODULA® WMS - PREMIUM



The MODULA® WMS – PREMIUM software module includes functionality designed to make the MODULA® LIFT VLM units as efficient as possible.

Main features of the MODULA® WMS – PREMIUM software module include:

- Dynamic item to compartment allocation management.
- Dynamic batching and item serial number management.
- FIFO and expiration date management.
- It also provides warehouse statistical analysis tools as well as print report customization.

The MODULA® WMS – PREMIUM software module adds the following features and functionalities to the MODULA® WMS – BASE software module:

- Item and location management with advanced features such as:
 - Free or tabled sub codes, material status management, packaging type management and expiration date management.
- Advanced user / operator management, it's possible to configure for each operator the enabled procedures required for their role.
- Customizable print reports and label reports management tools.
- Automatically assign an item to a preselected compartment size or location.
- An automatic and configurable import / export data exchange with a client's management system (host), using text files or shared tabled accessible via ODBC, .XML files, ASCII files, or using Excel files.
- Items can be managed with a FIFO sequence or on a sub code based priority.
- It is possible to retrieve statistical information of the VLM's movements and operations, as well as movement causes for immediate item requests.

MODULA® WMS – DRIVER & MODULA® LINK

For clients / users already having a "Host Management" type of software in place, the MODULA® WMS – DRIVER software module and/or MODULA® LINK allows the existing "Host" to communicate to and directly drive one or more MODULA® LIFT VLM units, with no need to develop direct communication protocols with the machines.

Using MODULA® WMS – DRIVER, users / operators will be able to still utilize the MODULA® LIFT VLM unit's CoPilot® operator console (HMI) for the required picking and replenishment activities as the information is clearly shown on the screen. Using MODULA® LINK it is possible to control the unit(s) with a PC during the loading/unloading operations and for picking confirmation, in place of the CoPilot® operator console (HMI). By purchasing the MODULA® WMS – DRIVER software module and/or MODULA® LINK, the customer is provided with the communications interface to perform an automatic data exchange.

Supplementary Modules Available For MODULA® WMS PREMIUM

NOTE: The supplementary software modules shown below are only available for purchase if the MODULA® WMS – PREMIUM software module has also been selected.

In addition to the MODULA® WMS PREMIUM software module, it is possible to add the following supplementary modules and increase both the WMS's features and an application's potential.



Advanced Item Management Module

The Advanced Item Management Module introduces the following features:

- Advanced security management, item and tray access can be limited depending on the user's role.
- It is possible to associate images to items.
- It is possible to group the items into families to define common characteristics.
- It is possible to group items into categories to improve search and filtering functionality.
- It is possible to manage supplementary attributes for items and orders.



Advanced Item Picking Module

The Advanced Item Picking Module extends the movement management features with the following features:

- Packing list management.
- Assembly kit management.
- Batch picking management, as well as the use of optional remote / external LED pick-to-light displays.
- Transit material management (boxes, totes, etc.).



Manual Warehouse Management Module

The Manual Warehouse Management Module allows for the creation and management of manual storage areas, in proximity to the MODULA® LIFT VLM's with the following features:

- Manual warehouse storage area management with a user friendly graphical tool.
- Picking management in manual areas.
- Material distribution between different areas based on accessed quantities.
- Material transfer between areas based on minimum stock values defined for each area.
- Incoming area management.

RF Module

The RF Module extends the functionality of the Manual Warehouse Management Module and it is only available in addition to the Manual Warehouse Management Module. This module allows the use RF terminals to manage manual storage areas.

NOTE:

The RF Module can only be purchased along with the Manual Warehouse Management Module. All RF terminals as well the Wi-Fi network coverage and network configuration must be provided by the customer.

SAP® IDOC Communication Module

Allows interfacing with the SAP® R3 WM module by means of IDOC exchange via FTP.

 MODULA® WMS Software Features/Functionality Overview

Functionalities:	BASE	AUTO	PREM	AI	AP	MW	RF	
Item List Management - Picking by Item No. or Order No.	☒	☒	☒	<i>Only Available in Addition To MODULA® WMS PREMIUM</i>				
Graphical Tray Location Management	☒	☒	☒					
Picking & Replenishment Order Management	☒	☒	☒					
Immediate Item or Order Request Management	☒	☒	☒					
Warehouse Inventory Management	☒	☒	☒					
Basic User Account Management - User Names & Passwords	☒	☒	☒					
Manual Data Import / Export via ASCII Files	☒	☒	☒					
Automatic Data Backup	☒	☒	☒					
Automatic Database Management	☒	☒	☒					
Movement History Management	☒	☒	☒					
Item Sub-Code Management			☒					
Tabled Item Sub-Code Management			☒					
Material Status Management			☒					
Packaging Management			☒					
Expiring Date Management			☒					
Advanced User Management			☒					
Customizable Reports			☒					
Customizable Label Creation Tool			☒					
Automatic Item to Compartment Allocation - Dynamic Allocation			☒					
Configurable Automatic Import / Export		☒	☒					
Item Management (Volume, FIFO, Priority)			☒					
Warehouse Statistics			☒					
Movement Cause Management for Immediate Requests			☒					
Item Based Security Management		☒		☒				
Items Management with Item Photos/Images				☒				
Manage Items by Family Type				☒				
Manage Items by Category Type				☒				
Manage Items by Their Supplementary Attributes				☒				
Manage Packing Lists for Items and Orders					☒			
Kiting Management					☒			
Item Storage Container & Order Tote Management					☒			
Management of Transit Materials					☒			
Management of Manual Warehouse Areas Apart from the VLM(s)						☒		
Management of Movement Between Areas						☒		
Incoming Area Management						☒		
RF Terminals Management							☒	

BASE: WMS BASE; **AUTO:** WMS AUTOMOTIVE; **PREM:** WMS PREMIUM

AI: Advanced Item; **AP:** Advanced Picking; **MW:** Manual Warehouse; **RF:** RF Module; **TS:** Remote Assistance

MODULA® LIFT VLM Third parts Software Integrations

DMS Integration Details:

Modula WMS for Automotive Dealerships Systems prepared for integration with the automotive **Dealer Management System (DMS)** includes a certified interface between Modula and the automotive dealer DMS software. DMS licensing and DMS fees includes one-time DMS installation, and fee costs from DMS provider to establish the certified interface for one (1) year.

To maintain the certified interface, the fees must be renewed annually to Modula Inc.

Modula will notify in advance when these fees are due with payment made directly to Modula. These yearly fees are subject to change based on current charges from the DMS Company to Modula.



DMS Integration

PC Requirements For The Server / Area Work Station PC

For all MODULA® WMS software, a PC must be supplied by the client's I.T. staff. The necessary hardware requirements are shown below:

MODULA® WMS Server / Work Station PC's (Minimum Requirements):

- MINIMUM REQUIRED CPU: Intel® Core™ i5-4690 (Quad-core, 3,50 GHz Turbo, 6 MB) - SUGGESTED CPU: Intel® Core™ i7-4790 (Quad-core HT, 3,60 GHz Turbo, 8 MB)
- MINIMUM REQUIRED RAM: 8 GB DDR3 1.600 MHz - SUGGESTED RAM: 16 GB DDR3 1.600 MHz
- MINIMUM REQUIRED STORAGE: HDD SATA da 3,5" 100 GB. (Suggested 2 HD of 100 GB in mirroring hardware RAID1) - SUGGESTED STORAGE: SDD SATA a 6 Gb/s da 128 GB (Suggested 2 SSD of 128GB in mirroring hardware RAID1)
- ETHERNET CONNECTIONS 10/100/1000 Mbit (RJ45): - Nr 2 ethernet card network card (mandatory). - 1 ethernet card connected to Host. - 1 ethernet card connected to Modula network configured with static IP address.
- IN/OUT: - DVD/CD-ROM Reader - USB keyboard and mouse - 1 USB 2.0 free port - Graphic card 1280X1024 75Hz (Suggested 1920X1080 75Hz) - Monitor 17", resolution 1280X1024 with Vga o Dvi cable (see your graphic card) (Suggested 1920X1080 75Hz)
- SUPPORTED OPERATING SYSTEMS: - 64 Bit CPU: - Windows 8.1 (Standard, Professional or Enterprise)* - Windows 10 (Professional, Enterprise)* - Windows Server 2012 R2 (Standard, DataCenter)* - Windows Server 2016 (Standard, DataCenter)* - - (*) The following windows features and function must be already enable\installed: - Framework 4.6.2 - Windows Installer 4.5 - Windows PowerShell - <u>All versions should be released with latest Windows Updates and Service Pack installed.</u>
- If connected to a company network data backup can be performed by the server through the network. - **If not, a back-up unit is required. Data back-up is mandatory. We suggest a 256 MB optic magnetic disk unit. Data backup is obligatory
An uninterrupted power supply for the PC is recommended.

PC Requirements For The Client

MODULA[®] WMS Client (Minimum Configuration):

- CPU: Intel [®] Core™ I3-4160 (3MB Cache, 3.60 GHz)
- RAM: 8 GB DDR3 1.600 MHz
- HARD DISK: At least 20 GB.
- ETHERNET CONNECTIONS 10/100/1000 Mbit (RJ45): - 1 ethernet card connected to Host.
- IN/OUT: - DVD/CD-ROM Reader - USB keyboard and mouse - 1 USB 2.0 free port - Graphic card 1280X1024 75Hz (Suggested 1920X1080 75Hz) - Monitor 17", resolution 1280X1024 with Vga o Dvi cable (see your graphic card) (Suggested 1920X1080 75Hz)
- SUPPORTED OPERATING SYSTEMS: <u>64 Bit CPU:</u> - Windows 7 x64 (Ultimate, Enterprise, Datacenter) - Windows 8.1 (Standard, Professional or Enterprise)* - Windows 10 (Professional, Enterprise)* <u>32 Bit CPU :</u> - Windows 7 (Ultimate, Enterprise, Datacenter) - Windows 8.1 (Standard, Professional or Enterprise)* - Windows 10 (Professional, Enterprise)* <u>(*) The following windows features and function must be already enable\installed:</u> - Framework 4.6.2 <u>All versions should be released with latest Windows Updates and Service Pack installed.</u>